

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

(1) 4 : 1

(2) 2 : 1

(3) 3 : 1

(4) 1 : 4

Q2: 24 Feb (Shift 1) - Numerical

An unpolarized light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 Lumens. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.

Q3: 24 Feb (Shift 2) - Single Correct

If the source of light used in a Young's double slit experiment is changed from red to violet:

(1) the fringes will become brighter.

(2) consecutive fringe lines will come closer.

(3) the central bright fringe will become a dark fringe.

(4) the intensity of minima will increase.

Q4: 25 Feb (Shift 1) - Single Correct

Two coherent light sources having intensity in the ratio $2 \times$ produce an interference pattern. The ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be :

(1) $\frac{2\sqrt{2x}}{x+1}$

(2) $\frac{\sqrt{2x}}{2x+1}$

Questions with Answer Keys

MathonGo

(3) $\frac{2\sqrt{2x}}{2x+1}$

(4) $\frac{\sqrt{2x}}{x+1}$

Q5: 25 Feb (Shift 2) - Single Correct

Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter $0.1\mu\text{m}$. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that:

- (1) its size decreases, but intensity increases
- (2) its size increases, but intensity decreases
- (3) its size increases, and intensity increases
- (4) its size decreases, and intensity decreases

Q6: 26 Feb (Shift 1) - Single Correct

In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away.

When a light of wavelength 500nm is used, the fringe separation will be :

- (1) 0.75 mm
- (2) 0.50 mm
- (3) 1 mm
- (4) 0.25 mm

Answer Key**Q1 (1)****Q2 (75)****Q3 (2)****Q4 (3)****Q5 (1)****Q6 (4)**

Q1 (1)

given : $\omega_2 = 3\omega_1$ also, $A \propto \omega$

$$\frac{\omega_1}{\omega_2} = \frac{1}{3} \dots (1)$$

Assume $\omega_1 = x, \omega_2 = 3x$

we know that

$$I_{\max} = (A_1 + A_2)^2, \text{ and}$$

$$I_{\min} = (A_1 - A_2)^2$$

$$\frac{A_1}{A_2} = \frac{\omega_1}{\omega_2} \dots (2)$$

from equation (2) we can say that

$$A_1 = A \text{ and } A_2 = 3A$$

$$\text{Now, } \frac{I_{\max}}{I_{\min}} = \frac{(A+3A)^2}{(A-3A)^2} = \frac{16A^2}{4A^2} = \frac{4}{1}$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{4}{1}$$

Q2 (75)

Given : $I_0 = 100$ lumens, $\theta = 30^\circ$

$$I_{\text{net}} = I_0 \cos^2 \theta$$

Hints and Solutions

MathonGo

$$I_{\text{net}} = 100 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{100 \times 3}{4}$$

$$I_{\text{net}} = 75 \text{ lumens}$$

Q3 (2)

$$\beta = \frac{\lambda D}{d}$$

$$\text{As } \lambda_v < \lambda_R$$

$$\Rightarrow \beta_v < \beta_R$$

$\therefore$ Consecutive fringe line will come closer.

$\therefore$ (2)

Q4 (3)

$$\text{Let } I_1 = 2x$$

$$I_2 = 1$$

$$I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \frac{(\sqrt{2x} + 1)^2 - (\sqrt{2x} - 1)^2}{(\sqrt{2x} + 1)^2 + (\sqrt{2x} - 1)^2}$$

$$= \frac{4\sqrt{2x}}{2+4x} = \frac{2\sqrt{2x}}{1+2x}$$

Q5 (1)

$$\sin \theta = \frac{1.22\lambda}{D}$$

If D is increased, then $\sin \theta$ will decreased

$\therefore$ size of circular fringe will decrease but intensity increases

Q6 (4)

$$\text{Fringe width } (\beta) = \frac{\lambda D}{d}$$

$$d = 2 \times 10^{-3} \text{ m}$$

$$\lambda = 500 \times 10^{-9} \text{ m}$$

$$D = 1 \text{ m}$$

Hints and Solutions

MathonGo

Now

$$\beta = \frac{500 \times 10^{-9} \times 1}{2 \times 10^{-3}}$$

$$\beta = \frac{5}{2} \times 10^{-4}$$

$$\beta = 2.5 \times 10^{-4}$$

$$\beta = 0.25 \text{ mm}$$